LAB # 1

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Lab Section : C

1. Introduction

The aim of this experiment is to learn how to use basic laboratory instruments such as the DC power supply, multimeter, breadboard, and ADALM2000 module. In this lab, simple resistor circuits are built in series and parallel, and voltage and current values are measured.

The experiment also aims to verify Ohm's Law and Kirchhoff's Laws through measurements. By applying different voltages and using different resistor values, the relationship between voltage, current, and resistance is observed. In addition, equivalent resistance of series and parallel resistor combinations is calculated and compared with measured results.

2. Pre-Lab Work

Before coming to the laboratory, I read the related sections in the lab manual about the DC power supply, multimeter, and breadboard. I also studied the ADALM2000 guidebook and reviewed its basic functions through the Scopy software.

Through these guidebooks, I learned the basic features and safe usage of the instruments that will be used in this experiment. I understood how to set voltage levels on the DC power supply, how to measure voltage, current, and resistance with the multimeter, and how connections are made on a breadboard. In addition, I reviewed the operating principles of the ADALM2000 module and its measurement channels.

3. Data/Results

Part 1 – Power Supply & Voltmeter – Workbench

Building Block:

In this part of the experiment, the DC power supply and multimeter were used to verify the accuracy of DC voltage measurements. The DC power supply output (+25 V terminal and common) was connected directly to the multimeter in DC voltage mode.

Components Used:

- DC Power Supply
- Multimeter
- Connection cables (alligator clips)

The positive terminal of the DC supply was connected to the positive input of the multimeter, and the common terminal was connected to the COM port of the multimeter. The circuit is a direct measurement setup where the multimeter is connected in parallel with the power supply output to measure the DC voltage.

Measurement:

The DC power supply voltage was set to different values, and the output was measured using the multimeter. The results are summarized below:

DC Power Supply	Voltmeter
0.5 V	0.499 V
2 V	1.998 V
5 V	4.999 V
10 V	10.002 V
15 V	14.996 V

Part 2 – Power Supply & Measurements – Workbench

Building Block

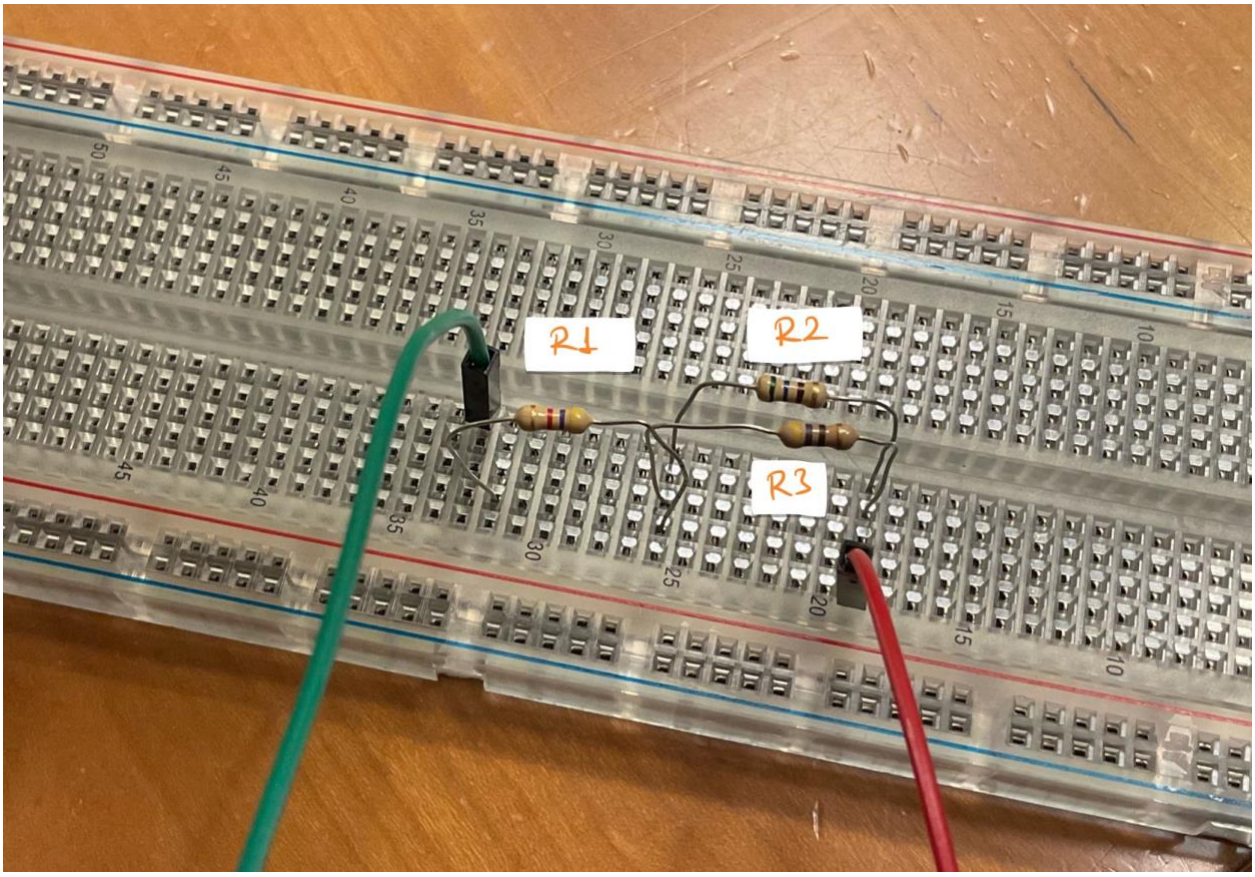
In this part of the experiment, three different resistors were selected and their resistance values were measured using the multimeter in Ω-mode. The nominal and measured values are listed below:

Resistor 1:
Nominal value: 1 kΩ
Measured value: 0.987 kΩ

Resistor 2:
Nominal value: 3.3 kΩ
Measured value: 3.276 kΩ

Resistor 3:
Nominal value: 7.5 kΩ
Measured value: 7.376 kΩ

After measurement, the resistors were labeled as R1, R2, and R3. R2 and R3 were connected in parallel on the breadboard. This parallel combination was then connected in series with R1.



The total resistance of the circuit was first calculated theoretically and then measured using the multimeter.

$R_{eq} = R1+(R2//R3) = 1000 + (3300 \times 7500)/(3300 + 7500)$

Calculated Total Resistance: 3.255 kΩ
Measured Total Resistance: 3.258 kΩ

The calculated and measured values are in very close agreement.

Measurement

A DC voltage of 4 V was applied across the entire circuit. The voltage drop across each resistor was measured using the multimeter in voltage mode.

Measured Voltage Drops:

$V_{R1} = 1.212$ V
$V_{R2} = 2.78$ V
$V_{R3} = 2.78$ V

As expected, the voltage across R2 and R3 is the same because they are connected in parallel. The voltage across R1 is different since it is connected in series with the parallel branch.Next, the current flowing through each resistor was measured using the multimeter in current mode.

Measured Currents:

$I_{R1} = 1.2$ mA
$I_{R2} = 0.828$ mA
$I_{R3} = 0.37$ mA

The measured results are consistent with circuit theory. Since R1 is in series with the parallel combination, the total current flows through R1. In the parallel branch, the current divides between R2 and R3 according to their resistance values. The resistor with lower resistance carries higher current,

which is consistent with the measurements.

- Calculate and measure the total resistance of the circuit using the multimeter (in Ω-mode).

Calculation Result: 3.255 kΩ Measurement Result: 3.258 kΩ

- Now use the power supply to apply 4 V DC across the circuit. Measure and record the potential drop across each resistor in turn using the multimeter (in V-mode). Are the measurement results as expected?

$V_{resistor1}$: 1.212 V $V_{resistor2}$: 2.78 V $V_{resistor3}$: 2.78 V

- Measure and record the currents flowing over each resistor using the multimeter (in I-mode). Note that you have to switch the cable configuration for this measurement. Look at the lab manual!

$I_{resistor1}$: 1.2 mA $I_{resistor2}$: 0.828 mA $I_{resistor3}$: 0.37 mA

Part 3 – Power Supply & Voltmeter – ADALM2000

Building Block

In this part of the experiment, the ADALM2000 module and Scopy software were used as a DC power supply and voltmeter. The module was connected to the computer via USB, and the Power Supply and Voltmeter tabs in Scopy were used to generate and measure DC voltages.

Components / Instruments Used:
• ADALM2000 module
• Computer with Scopy software
• Jumper cables

The positive and ground outputs of the ADALM2000 power supply were connected with jumper cables. Then, Channel 1 of the voltmeter (DC configuration) was used to measure the output voltage directly across these two jumper cables.

Measurement

First, the output voltage was set to 2.5 V and the voltage was checked using the Scopy voltmeter. The measured value was close to the set value, confirming correct setup.

Next, the power supply was adjusted between 0 V and +5 V, and the voltmeter readings were recorded. Error percentages were calculated and listed in the table:

Power Supply (V) → Voltmeter (V) → Error (%)

Power Supply (V)	Voltmeter (V)	Error Percentage (%)
0.1	0.095 V	%.5
0.25	0.244 V	%.2.4
0.5	0.497 V	%.0.6
1	0.997 V	%.0.3
5	4.944 V	%.1.12

The results show that the ADALM2000 voltmeter readings are generally very close to the set voltage values, especially at higher voltages. The largest relative error was observed at the lowest setting (0.1 V), which is expected due to resolution and accuracy limits at small voltage levels.

Part 4 – Ohm's Law – ADALM2000

Building Block

In this part of the experiment, Ohm's Law was verified using an LED and different series resistors powered by the 5 V output of the ADALM2000 module.

Components Used:
- ADALM2000 (5 V supply)
- LED (D1)
- 470 Ω resistor
- 4.7 kΩ resistor
- 24 kΩ resistor
- Jumper cables

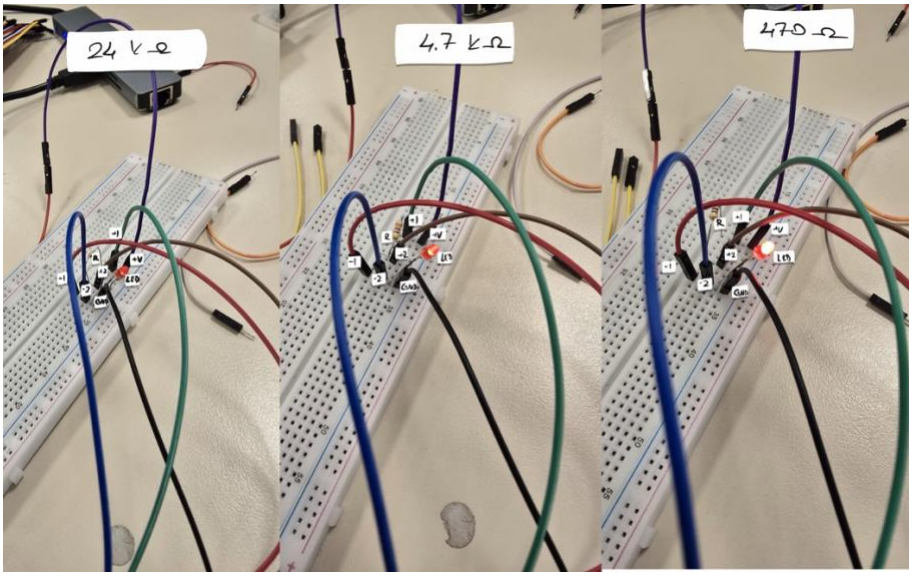
The LED and resistor were connected in series. Channel 1 of the ADALM2000 was used to measure the supply voltage (+5 V), and Channel 2 was used to measure the voltage across the LED. The resistor voltage (VR) was determined by subtracting the LED voltage from the supply voltage:

$VR = Vs - VLED$

The current flowing through the circuit was calculated using Ohm's Law:

$I = VR / R$

The experiment was repeated using three different resistor values (470 Ω, 4.7 kΩ, and 24 kΩ) to observe changes in LED brightness.



Measurement

For each resistor value, the LED brightness and calculated current were compared.

1. R = 24 kΩ
   The LED was very dim or barely visible. With a large resistance, the current became very small. Since LED brightness is directly related to current, the light intensity significantly decreased.

2. R = 4.7 kΩ
   The LED brightness was moderate. Increasing the resistance reduced the current flowing through the circuit. As a result, the LED emitted less light compared to the 470 Ω case.

3. R = 470 Ω
   The LED was very bright. Since the resistance is small, the voltage drop across the resistor is lower, resulting in a higher current. According to Ohm's Law, a smaller resistance leads to a larger current. This high current caused the LED to glow strongly.

The results clearly demonstrate the relationship between current and LED brightness. As predicted by Ohm's Law, increasing the resistance decreases the current, which reduces the brightness of the LED. This experiment confirms that current is inversely proportional to resistance when the supply voltage is constant.

4. Discussion/Analysis

In this experiment, the theoretical calculations, instrument settings, and experimental measurements were compared to evaluate the validity of Ohm's Law and the behavior of series and parallel resistor combinations.

In Part 1 and Part 3, the voltage values set on the DC power supply (both workbench supply and ADALM2000) were compared with the voltmeter readings. The measured values were very close to the set values in all cases. Small deviations were observed, especially at lower voltage levels, but these differences were within acceptable measurement tolerance. This indicates that both the DC power supply and the measuring instruments operate reliably and consistently.

In Part 2, the calculated total resistance (3.255 kΩ) and the measured total resistance (3.258 kΩ) were almost identical. The voltage drops across the resistors also behaved as expected: resistors connected in parallel had the same voltage drop, and the series resistor had a different voltage consistent with circuit theory. The measured currents confirmed that the total current flows through the series element and divides in the parallel branch according to resistance values. These results are consistent with Ohm's Law and Kirchhoff's Laws.

In Part 4, the relationship between current and LED brightness was clearly observed. When a smaller resistor (470 Ω) was used, the LED was very bright due to higher current. As the resistance increased (4.7 kΩ and 24 kΩ), the current decreased and the LED brightness reduced significantly. This confirms that current is inversely proportional to resistance when the supply voltage is constant.

Although the results were very close to theoretical expectations, minor differences between calculated and measured values may be due to several factors. These include resistor tolerance, internal resistance of measurement devices, contact resistance on the breadboard, and small fluctuations in the power supply output. In addition, LEDs are non-ideal components with nonlinear voltage-current characteristics, which may also introduce slight deviations from ideal Ohm's Law behavior.

5. Post-Lab

A voltmeter is connected in parallel with a resistor because voltage is defined as the potential difference between two points. To measure the voltage across a resistor correctly, the voltmeter must be placed across its terminals. Ideally, a voltmeter has infinite internal resistance, so it does not draw current from the circuit. Therefore, the ideal current flowing through a voltmeter is zero.

An ampermeter is connected in series with the resistor because it must measure the current flowing through that component. Since current is the same throughout a series path, the ammeter must be

inserted directly into the circuit path. Ideally, an ammeter has zero internal resistance so that it does not affect the circuit. In the ideal case, the voltage drop across an ammeter is zero.

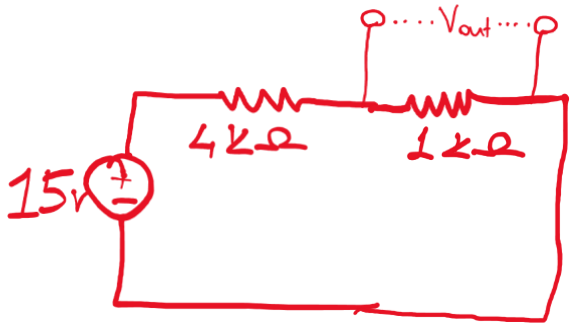
If only a +15 V source is available but +3 V is required, a simple voltage divider can be designed using two resistors in series. To obtain 3 V from 15 V, the output must be one-fifth of the input. Therefore, the resistor ratio should satisfy:

Vout = (R2 / (R1 + R2)) × 15 V = 3 V

This requires R2 / (R1 + R2) = 1/5, meaning R1 should be four times R2. For example, R1 = 4 kΩ and R2 = 1 kΩ would produce approximately 3 V at the midpoint.

The main problem with this circuit is how it reacts when you actually try to use it. When you connect a device to the 3V output, that device has its own internal resistance. This creates a parallel connection with your second resistor, which immediately lowers the total resistance of the bottom part of your circuit.

Because voltage is shared based on the size of the resistors, making the bottom resistance smaller means it can no longer hold onto its share of the 15V. Most of the voltage will stay trapped at the top resistor, and your 3V output will drop down to a much lower level, like 1V or 2V. This means your device won't get the power it needs to stay on. Also the resistors are constantly wasting energy by turning it into heat, which makes this a very inefficient way to power anything that needs a lot of electricity.



6. Conclusion

In this experiment, we had the opportunity to work directly with basic circuit components and measurement instruments, which helped reinforce the theoretical concepts learned in class. The measured voltage, current, and resistance values were very close to the calculated results, showing that Ohm's Law and fundamental circuit principles accurately describe real circuit behavior.

We also gained practical insight into how measurement devices should be connected. Understanding why the voltmeter must be connected in parallel and the ammeter in series made the theory more meaningful, since we could see how incorrect connections would affect the results. Additionally, small differences between theoretical and measured values reminded us that real components are not ideal and that internal resistance and tolerances always exist in practical systems.